

Computational physics

Partial Differential Equations

1

Partial Differential Equations

- Involving more than one independent variable
- Physics problems → second-order PDEs
- elliptic, parabolic, or hyperbolic

2

PDEs in physics

- General form

$$A \frac{\partial^2 U}{\partial x^2} + 2B \frac{\partial^2 U}{\partial x \partial y} + C \frac{\partial^2 U}{\partial y^2} + D \frac{\partial U}{\partial x} + E \frac{\partial U}{\partial y} = F$$

- Elliptic: $AC - B^2 > 0$

- Laplace's equation:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \equiv \nabla^2 u(x, y) \equiv \Delta u(x, y) = 0$$

- Poisson's equation:

$$\nabla^2 u(x, y) = -4\pi\rho(x, y)$$

3

PDEs in physics

- Parabolic: $AC - B^2 = 0$

- Heat equation:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \nabla^2 u(x, y, t) = \alpha \frac{\partial u}{\partial t}$$

- Hyperbolic: $AC - B^2 < 0$

- Wave equation:

$$\nabla^2 u(x, y, t) = -c^2 \frac{\partial^2 u}{\partial t^2}$$

4

Numerical solutions to elliptic equations

- Elliptic equation

$$\nabla^2 u(x, y) \equiv \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$$

- Taylor expansions of the potential $u(x, y)$ to the right and left of (x, y)

$$u(x - \Delta x, y) = u(x, y) - \frac{\partial u}{\partial x} \Delta x + \frac{1}{2} \frac{\partial^2 u}{\partial x^2} (\Delta x)^2 - \dots [1]$$

$$u(x + \Delta x, y) = u(x, y) + \frac{\partial u}{\partial x} \Delta x + \frac{1}{2} \frac{\partial^2 u}{\partial x^2} (\Delta x)^2 + \dots [2]$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} \approx \frac{u(x + \Delta x, y) + u(x - \Delta x, y) - 2u(x, y)}{(\Delta x)^2}$$

5

Numerical solutions to elliptic equations

- Taylor expansions of the potential $u(x, y)$ to the right and left of (x, y)

$$u(x, y - \Delta y) = u(x, y) - \frac{\partial u}{\partial y} \Delta y + \frac{1}{2} \frac{\partial^2 u}{\partial y^2} (\Delta y)^2 - \dots$$

$$u(x, y + \Delta y) = u(x, y) + \frac{\partial u}{\partial y} \Delta y + \frac{1}{2} \frac{\partial^2 u}{\partial y^2} (\Delta y)^2 + \dots$$

- Adding the two series:

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} \approx \frac{u(x, y + \Delta y) + u(x, y - \Delta y) - 2u(x, y)}{(\Delta y)^2}$$

6

Numerical solutions to elliptic equations

$$\frac{\partial^2 u}{\partial x^2} \approx \frac{u(x + \Delta x, y) + u(x - \Delta x, y) - 2u(x, y)}{(\Delta x)^2}$$

$$\frac{\partial^2 u}{\partial y^2} \approx \frac{u(x, y + \Delta y) + u(x, y - \Delta y) - 2u(x, y)}{(\Delta y)^2}$$

- Adding the two equations:

$$\Rightarrow \frac{u(x + \Delta x, y) + u(x - \Delta x, y) - 2u(x, y)}{(\Delta x)^2} + \frac{u(x, y + \Delta y) + u(x, y - \Delta y) - 2u(x, y)}{(\Delta y)^2} = f(x, y)$$

7

Numerical solutions to elliptic equations

- For $\Delta x \equiv h, \Delta y = k$:

$$\begin{aligned} & \frac{u(x + h, y) + u(x - h, y) - 2u(x, y)}{h^2} + \\ & \frac{u(x, y + k) + u(x, y - k) - 2u(x, y)}{k^2} = f(x, y) \\ \Rightarrow & 2 \left[\frac{h^2}{k^2} + 1 \right] u(x, y) - [u(x + h, y) + u(x - h, y)] \\ & - \frac{h^2}{k^2} [u(x, y + k) + u(x, y - k)] = -h^2 f(x, y) \end{aligned}$$

8

Numerical solutions to elliptic equations

- For $\Delta x = \Delta y = h$:

$$\begin{aligned} & 4u(x, y) - [u(x + h, y) + u(x - h, y)] \\ & - [u(x, y + h) + u(x, y - h)] = -h^2 f(x, y) \\ \Rightarrow & 4u(x, y) \approx [u(x + h, y) + u(x - h, y)] \\ & + [u(x, y + h) + u(x, y - h)] - h^2 f(x, y) \end{aligned}$$

- Poisson's equation:

$$\begin{aligned} & 4u(x, y) \approx u(x + h, y) + u(x - h, y) \\ & + u(x, y + h) + u(x, y - h) + h^2 4\pi\rho(x, y) \end{aligned}$$

9

Numerical solutions to elliptic equations

- Finite-Difference method: $x \rightarrow x_i = x_0 + ih$,

$$y \rightarrow y_j = y_0 + jk ; i = 0, \dots, n, j = 0, \dots, m$$

$$\begin{aligned} \Rightarrow & 2 \left[\frac{h^2}{k^2} + 1 \right] u_{ij} - [u_{i+1,j} + u_{i-1,j}] - \frac{h^2}{k^2} [u_{i,j+1} + u_{i,j-1}] \\ = & -h^2 f(x_i, y_j) \end{aligned}$$

- with $u_{ij} \equiv u(x_i, y_j)$; $i = 1, \dots, n - 1$, $j = 1, \dots, m - 1$
- And boundary conditions

$$u_{0j} = g(x_0, y_j), u_{nj} = g(x_n, y_j), j = 0, \dots, m$$

$$u_{i0} = g(x_i, y_0), u_{im} = g(x_i, y_m), i = 0, \dots, n$$

10

Numerical solutions to elliptic equations

Poisson's equation [$h = k$]

$$u_{ij} = \frac{1}{4} [u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1}] + h^2 \pi \rho(x_i, y_j)$$

Laplace's equation [$h = k$]

$$u_{ij} = \frac{1}{4} [u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1}]$$

- With boundary conditions

$$u_{0j} = g(x_0, y_j), u_{nj} = g(x_n, y_j), j = 0, \dots, m$$

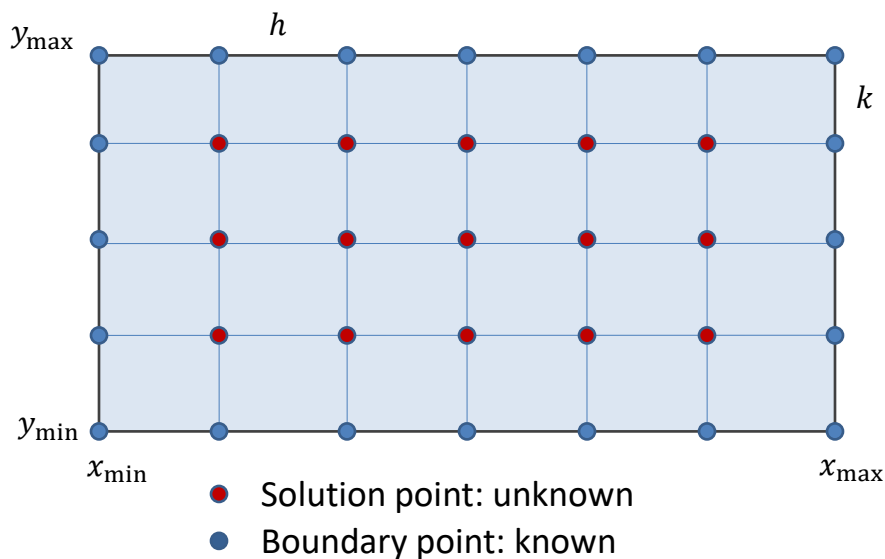
$$u_{i0} = g(x_i, y_0), u_{im} = g(x_i, y_m), i = 0, \dots, n$$

- This Poisson/Laplace's equation is equivalent to a system of linear equations! We use the Gauss-Seidel method to solve it.

11

Numerical solutions to elliptic equations

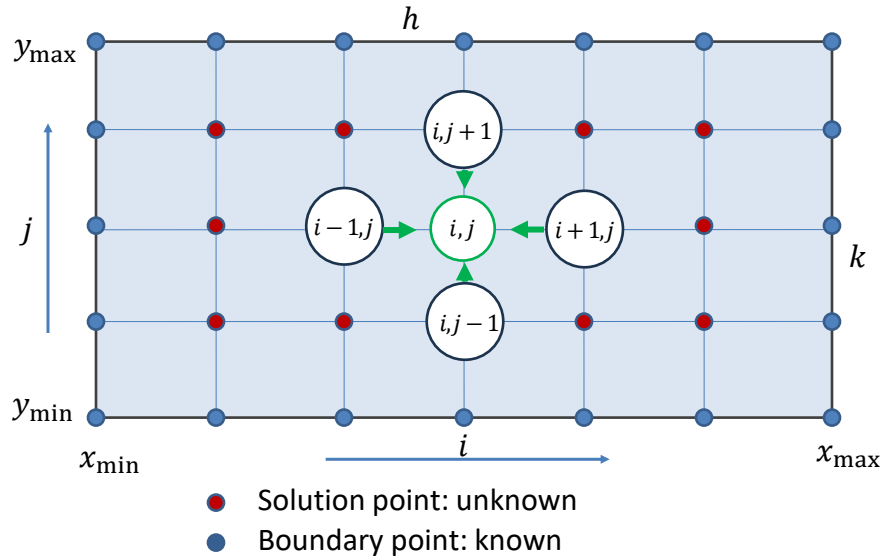
$$u_{ij} = \frac{1}{4} [u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1}] + h^2 \pi \rho(x_i, y_j)$$



12

Numerical solutions to elliptic equations

$$u_{ij} = \frac{1}{4} [u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1}] + h^2 \pi \rho(x_i, y_j)$$



13

Jacobi iterative method

$$u_{ij} = \frac{1}{4} [u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1}] + h^2 \pi \rho(x_i, y_j)$$

- The Jacobi iterative method is obtained by solving the i -th equation in $A\mathbf{x} = \mathbf{b}$ for x_i to obtain (provided $a_{ii} \neq 0$)

$$x_i = \frac{1}{a_{ii}} \left[\sum_{j=1, j \neq i}^n -a_{ij}x_j + b_i \right], i = 1, 2, \dots, n$$

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n \end{cases} \Rightarrow x_i = \frac{1}{a_{ii}} \left[\sum_{j=1, j \neq i}^n -a_{ij}x_j + b_i \right]$$

14

Jacobi iterative method

- For each $k \geq 1$, generate the components $x_i^{(k)}$ ["new"] of $\mathbf{x}^{(k)}$ from the components of $\mathbf{x}^{(k-1)}$ ["old" / known/ đã biết/ đã có] by

$$x_i^{(k)} = \frac{1}{a_{ii}} \left[\sum_{\substack{j=1 \\ j \neq i}}^n -a_{ij} x_j^{(k-1)} + b_i \right], i = 1, 2, \dots, n$$

$$x_i^{(\text{new})} = \frac{1}{a_{ii}} \left[\sum_{\substack{j=1 \\ j \neq i}}^n -a_{ij} x_j^{(\text{old})} + b_i \right], i = 1, 2, \dots, n$$

15

Gauss-Seidel method

$$u_{ij} = \frac{1}{4} [u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1}] + h^2 \pi \rho(x_i, y_j)$$

- Jacobi's method

$$x_i^{(k)} = \frac{1}{a_{ii}} \left[- \sum_{j=1}^n a_{ij} x_j^{(k-1)} + b_i \right], i = 1, 2, \dots, n$$

- Gauss-Seidel method

$$x_i^{(k)} = \frac{1}{a_{ii}} \left[- \sum_{j=1}^{i-1} a_{ij} x_j^{(k)} - \sum_{j=i+1}^n a_{ij} x_j^{(k-1)} + b_i \right]$$

16

Gauss-Seidel Iterative Algorithm

- Step 1: Set $k = 1$ // k : k -th iteration [lần lặp thứ k]
- Step 2: While ($k \leq N$) do Step 3 -6
 - Step 3: For $i = 1, n$

$$\text{Set } x_i = \frac{1}{a_{ii}} \left[-\sum_{j=1}^{i-1} a_{ij} x_j - \sum_{j=i+1}^n a_{ij} x_j + b_i \right]$$
 - Step 4: If $\|x - x^{\text{old}}\| < \epsilon$ then OUTPUT $x_1, \dots, x_n$
STOP
 - Step 5: Set $k = k + 1$
 - Step 6: For $i = 1, n$ set $x_i^{\text{old}} = x_i$
- Step 7: OUTPUT “Maximum number of iterations exceeded!”
STOP (The procedure was unfortunately unsuccessful ☹)

17

Numerical solutions to elliptic equations

- Sử dụng PP Gauss-Seidel cho PT Poisson:

$$u_{ij} = \frac{1}{4} [u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1}] + h^2 \pi \rho(x_i, y_j)$$

- for $i = 1, \dots, n - 1 ; j = 1, \dots, m - 1$
- Điều kiện biên

$$\begin{aligned} u_{0j} &= g(x_0, y_j), u_{nj} = g(x_n, y_j), j = 0, \dots, m \\ u_{i0} &= g(x_i, y_0), u_{im} = g(x_i, y_m), i = 0, \dots, n \end{aligned}$$

18

Numerical solutions to elliptic equations

- Sử dụng PP Gauss-Seidel cho PT Poisson:

$$u_{ij}^{(new)} = \frac{1}{4} \left[u_{i+1,j}^{(old)} + u_{i-1,j}^{(new)} + u_{i,j+1}^{(old)} + u_{i,j-1}^{(new)} \right] + h^2 \pi \rho(x_i, y_j)$$

- Or

$$u_{ij}^{(k)} = \frac{1}{4} \left[u_{i+1,j}^{(k-1)} + u_{i-1,j}^{(k)} + u_{i,j+1}^{(k-1)} + u_{i,j-1}^{(k)} \right] + h^2 \pi \rho(x_i, y_j)$$

- Điều kiện biên

$$u_{0j} = g(x_0, y_j), u_{nj} = g(x_n, y_j), j = 0, \dots, m$$

$$u_{i0} = g(x_i, y_0), u_{im} = g(x_i, y_m), i = 0, \dots, n$$

19

Numerical solutions to elliptic equations

- Sử dụng PP Gauss-Seidel cho PT Laplace:

$$u_{ij}^{(new)} = \frac{1}{4} \left[u_{i+1,j}^{(old)} + u_{i-1,j}^{(new)} + u_{i,j+1}^{(old)} + u_{i,j-1}^{(new)} \right]$$

- Or

$$u_{ij}^{(k)} = \frac{1}{4} \left[u_{i+1,j}^{(k-1)} + u_{i-1,j}^{(k)} + u_{i,j+1}^{(k-1)} + u_{i,j-1}^{(k)} \right]$$

- Điều kiện biên

$$u_{0j} = g(x_0, y_j), u_{nj} = g(x_n, y_j), j = 0, \dots, m$$

$$u_{i0} = g(x_i, y_0), u_{im} = g(x_i, y_m), i = 0, \dots, n$$

20

Electrostatic Potentials

- “Find the electric potential for all points *inside the charge-free* square shown in the above Fig. The bottom and sides of the region are made up of wires that are “grounded” (kept at 0 V). The top wire is connected to a battery that keeps it at a constant 100V.”



21

G-S Iterative Algorithm for Electrostatic Potentials/ Laplace Equations

$$u_{ij}^{(k)} = \frac{1}{4} [u_{i+1,j}^{(k-1)} + u_{i-1,j}^{(k)} + u_{i,j+1}^{(k-1)} + u_{i,j-1}^{(k)}]$$

Step 1: Input N [Maximum number of iterations], n, m [nếu $n \neq m$];
 Bound conditions $u_{i,0} [i = 0, n - 1], u_{0,j} [j = 0, m - 1]; u_{i,n}, u_{m,j};$
 and $u_{ij}^{(0)} [i = 1, n - 2, j = 1, m - 2].$

S2: For $k = 0, N - 1$ // vòng lặp/iteration [thay cho “While”][do Step 3]

S3: For $i = 1, n - 2$ // hướng x: x-direction

For $j = 1, n - 2$ // hướng y. Nếu $m \neq n$ thì cần cho $j = 1, m - 2$

Set $u_{i,j} = (u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1})/4$

S4: // OUTPUT

For $i = 0, n - 1$ // x:

For $j = 0, n - 1$ // y. nếu $m \neq n$ thì cần cho $j = 1, m$

Output $x_0 + i * h, y_0 + j * h, u_{i,j}$

...

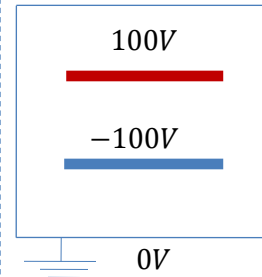
STOP

Nếu muốn chương trình dừng lại khi đạt được độ chính xác TOL $[\varepsilon]$
 thì cần thay đổi những gì trong code?

22

Thực hành

1. Lập trình giải bài toán thế tĩnh điện vừa xét ở slides trước và vẽ thế tĩnh điện. Ma trận A của hệ PT ứng với phương trình Laplace có dạng thế nào?
2. Hiệu chỉnh code để dừng khi đạt được độ chính xác ε , và cho biết lần lặp thứ k tương ứng với độ chính xác này. [Nội dung PP G-S đã xét đến.]
3. Giải số và vẽ thế cho bài toán ở hình bên:
2 thanh/sợi (dài bằng nhau) với điện thế 100V và -100V, đặt bên trong sợi hình vuông nối đất. Tính và vẽ điện thế bên trong sợi hình vuông. [Tự 'set' chiều dài thanh, khoảng cách giữa 2 thanh.]



26

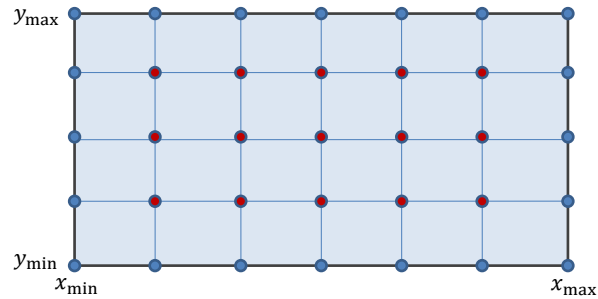
Thực hành

4. Giải PT Poisson cho thế tĩnh điện tạo ra bởi 1 điện tích điểm [vd, electron] được đặt tại giữa "box" [2D]. Cần xác định điều kiện biên như thế nào? Chọn độ chính xác, vd, $\varepsilon = 10^{-6}$; thử tăng giảm ε và quan sát số lần lặp tương ứng. Vẽ thế tĩnh điện của điện tích điểm.

27

Numerical solutions to elliptic equations

$$u_{ij} = \frac{1}{4} [u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1}] + h^2 \pi \rho(x_i, y_j)$$



- Hệ PT tuyến tính $(n - 1) \times (n - 1)$ [chọn $n = m$].
- Ma trận của hệ này "strictly diagonally dominant" không?